Rohit Kumar

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Research Interests

Decision-reliable machine learning: uncertainty that stays calibrated when data arrive late and shift, judging models by the decisions they drive, and knowing when adaptivity pays. Main testbed: power systems and electricity markets.

Education

M.Tech, Artificial Intelligence 2025 – May 2027 (expected)
Dr B R Ambedkar National Institute of Technology Jalandhar, India. Advisor: Dr Urvashi Bansal
CGPA 8.36/10. Research: forecasting and decision making for India's power system from public data.

B.Tech, Electrical and Electronics Engineering 2020 – 2024
International Institute of Information Technology (IIIT) Bhubaneswar, India
CGPA 7.56/10.

Manuscripts Under Review

1. **R. Kumar**, U. Bansal. Does adaptive risk aversion pay? A pre-registered evaluation of probabilistic day-ahead drawal scheduling under India's deviation settlement. *Under review, IEEE Transactions on Power Systems*, 2026.
2. **R. Kumar**, U. Bansal. When does adaptive correction of day-ahead forecasts pay? Evidence from India's power system. *Manuscript for Energy Conversion and Management*, 2026.

Publications

3. D. P. Mishra, S. Samal, **R. Kumar**, A. K. Sahoo, S. R. Salkuti. Renewable energy optimization for sustainable power generation. *International Journal of Informatics and Communication Technology*, 15(1):365–373, 2026. doi:10.11591/ijict.v15i1.pp365-373
4. D. P. Mishra, **R. Kumar**, A. K. Sahoo, T. Parida. Arc fault detection for improved electrical safety with ADC and relay. *IEEE International Conference on Advances in Power, Signal, and Information Technology (APSIT)*, 2023. IEEE Xplore 10201793

Dataset and Software

- **R. Kumar**. *RE-FUSED: State-day panel of India's power system, code and frozen results.* Zenodo, 2026. doi:10.5281/zenodo.23301898. Sole creator. 104,550 State-days × 107 columns, 28 States and 6 Union Territories, rebuilt by script from official reports and checked with SHA-256 hashes. Code: github.com/Ohi-Me/RE-FUSED

Research Experience

M.Tech research: forecasting and scheduling for India's power system 2025 – present
Centre for AI, NIT Jalandhar (advisor Dr Urvashi Bansal)

- Forecast five grid targets a day ahead using only data as published. Ensembles of gradient-boosted trees, deep networks and Chronos-2 beat the seasonal baseline on all 5 targets over 516 held-out days.
- Froze a pre-registration (52 checks, hash recorded) before opening the test period: 35 supported, the rest reported.
- Scheduling: in the paper's daily settlement model, a fixed quantile of a calibrated forecast cut cost from 18.54 to 15.80 crore rupees per day versus repeating the latest schedule. No adaptive or reinforcement-learning rule beat it. Proved the expected regret is convex in the schedule and reproduced all 302,400 official 15-minute charges (six control areas) to within 2.5 rupees.

- Forecast correction: a measurable condition for when adaptive correction pays, with two validation-only selection rules. Pre-registered tests: 19 of 23 supported on earlier benchmarks, 5 of 20 on India's hourly data.
- Audit habit: found that 98.2% of test-period prices in an earlier panel were carried forward, and withdrew the price-forecasting claim built on them.
- All experiments run as scheduled jobs on the institute's H100 GPU cluster.

Research proposal, Research Scheme on Power 2026
Central Power Research Institute (CPRI), Ministry of Power, India

- Co-wrote a 36-month, Rs 82.64 lakh proposal on weather-aware forecasting, uncertainty quantification and decision support for Indian power systems, and handled its technical work. Submitted, decision pending.

Awards and Competitions

- **Amazon ML Challenge 2026**: rank 206 of 89,000+ participants (team), 16th after day 1. Matched 1.7M businesses to about 10M records (gradient-boosted trees, graph neural network, fine-tuned XLM-R).
- **Kaggle RSNA Knee Abnormality Detection 2026**: rank 542 of 6,425 teams (top 9%). Labelled 4,407 MRI studies from radiology reports with an open LLM, checked against 58 expert labels.
- 2nd place, Swing 'n' Code programming contest, IIIT Bhubaneswar.
- Competitive programming: Codeforces 1388 (max), LeetCode 1663, CodeChef 1703.

Selected Projects

- **DataSentry**: data-reliability system. Split-conformal alert limits (observed false-positive rate 0.0104 against a 0.01 target), column-level lineage parsed from SQL, 115 tests. github.com/Ohi-Me/DataSentry
- **Orbit**: multi-agent (LangGraph) financial research platform with look-ahead-leakage tests and purged walk-forward validation. github.com/Ohi-Me/Orbit
- **FeelFit**: lab-report understanding (OCR, LOINC mapping, retrieval with citations), live on web and mobile. feelfit.vercel.app

Skills

Programming: Python, C++, SQL · **ML:** PyTorch, Hugging Face Transformers, XGBoost, scikit-learn · **Methods:** probabilistic and quantile forecasting, conformal prediction, walk-forward validation, pre-registration, significance testing (Diebold–Mariano) · **Tools:** Git, Docker, FastAPI, PBS job scheduling on GPU clusters

Service and Languages

Internship Representative, Institute Relations, NIT Jalandhar (2026 – present). Core member, Advaita tech fest (Photogeeks), IIIT Bhubaneswar (2023). Languages: English (fluent), Hindi (native).

References

Dr Urvashi Bansal, Centre for Artificial Intelligence, NIT Jalandhar
Prof. Sukumar Nandi, Department of Computer Science and Engineering, IIT Guwahati
Dr Debani Prasad Mishra, Department of Electrical and Electronics Engineering, IIIT Bhubaneswar